

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

CS 498: Inference Overview

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The Grainger College of Engineering

Materials with thanks to Minjia Zhang, Arvind Krishnamurthy, and many colleagues

Recall: Distributed Model Training

- **Model execution exceeds the capacity of single machines**
  - Data parallelism, pipeline parallelism, tensor parallelism

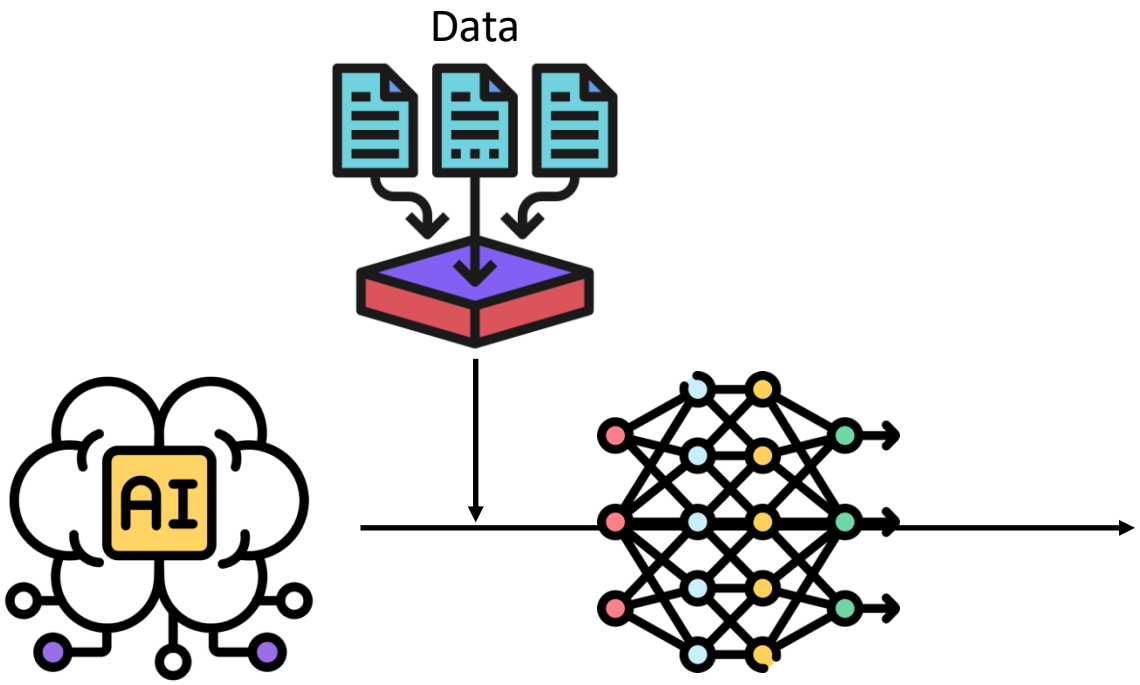
- **Reduce the resource demand of training**
  - Mixed precision training (e.g., INT8)
  - Gradient accumulation
  - Activation checkpointing
  - Low-rank adaptation

Today

LLM Inference Overview

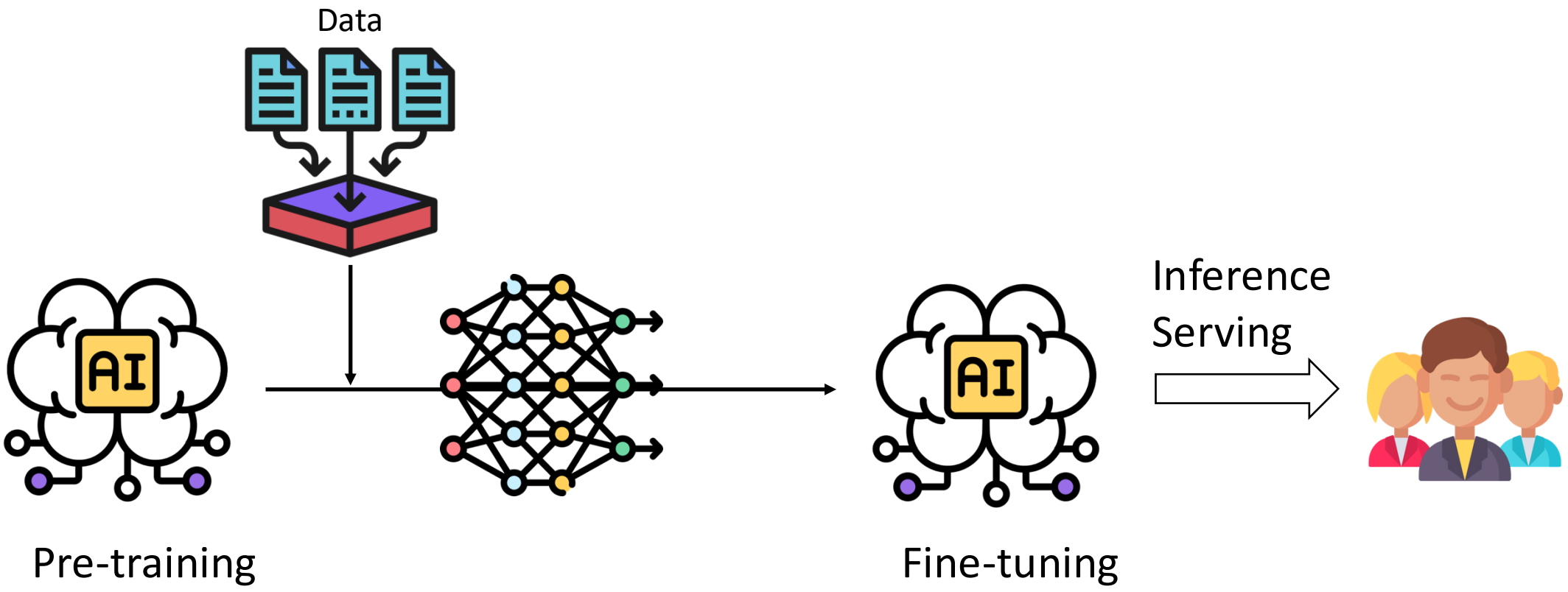
- Background
- Challenges
- System optimizations
- Algorithm optimizations

What is Model Inference/Serving?



Pre-training

What is Model Inference/Serving?



Inference Scenario 1: ChatGPT/SearchGPT/Copilot

What are some fun places to visit at Urbana Champaign?

Searching for: **fun places to visit at Urbana Champaign**

Generating answers for you...

There are many fun places to visit at Urbana Champaign, depending on your interests and preferences. Here are some of the most popular ones:

- If you love nature and animals, you might enjoy visiting the **Anita Purves Nature Center** [1], where you can explore trails, feed goats, and learn about local wildlife.
- If you are into art and culture, you might want to check out the **Krannert Art Museum** [2], which has a diverse collection of artworks from various regions and periods, as well as exhibitions and events.
- If you are looking for some entertainment and history, you might like the **Virginia Theatre** [3], which is a restored historic venue that hosts concerts, movies, and shows.
- If you are feeling adventurous and sporty, you might have fun at the **University of Illinois Ice Arena**, where you can skate, play hockey, or watch games.

These are just some of the fun places to visit at Urbana Champaign. You can find more information and reviews on Tripadvisor or Bing. I hope you have a great time exploring the city! 😊

Learn more 1 experiencecu.org 2 kam.illinois.edu 3 bing.com

2 of 30



Microsoft Copilot: Your AI companion

Inference Scenario 2: Online Image/Video Generation



DALL·E 3 · An expressive oil painting of a basketball player dunking, depicted as an explosion of a nebula.

[DALL·E 3 | OpenAI]



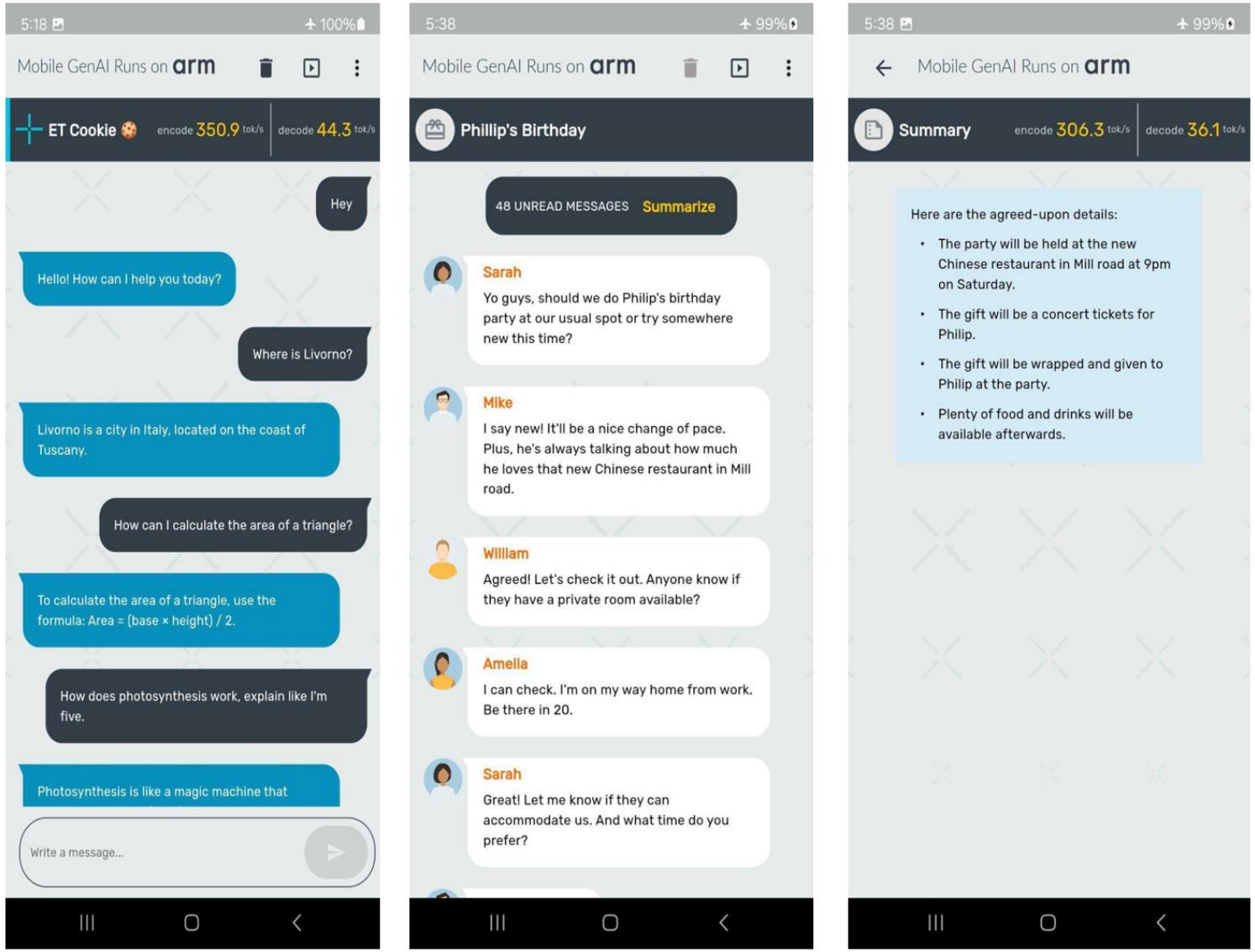
Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.

[Sora2: Creating video from text]

Inference Scenario 3: Surveillance System



Inference Scenario 4: Mobile Apps

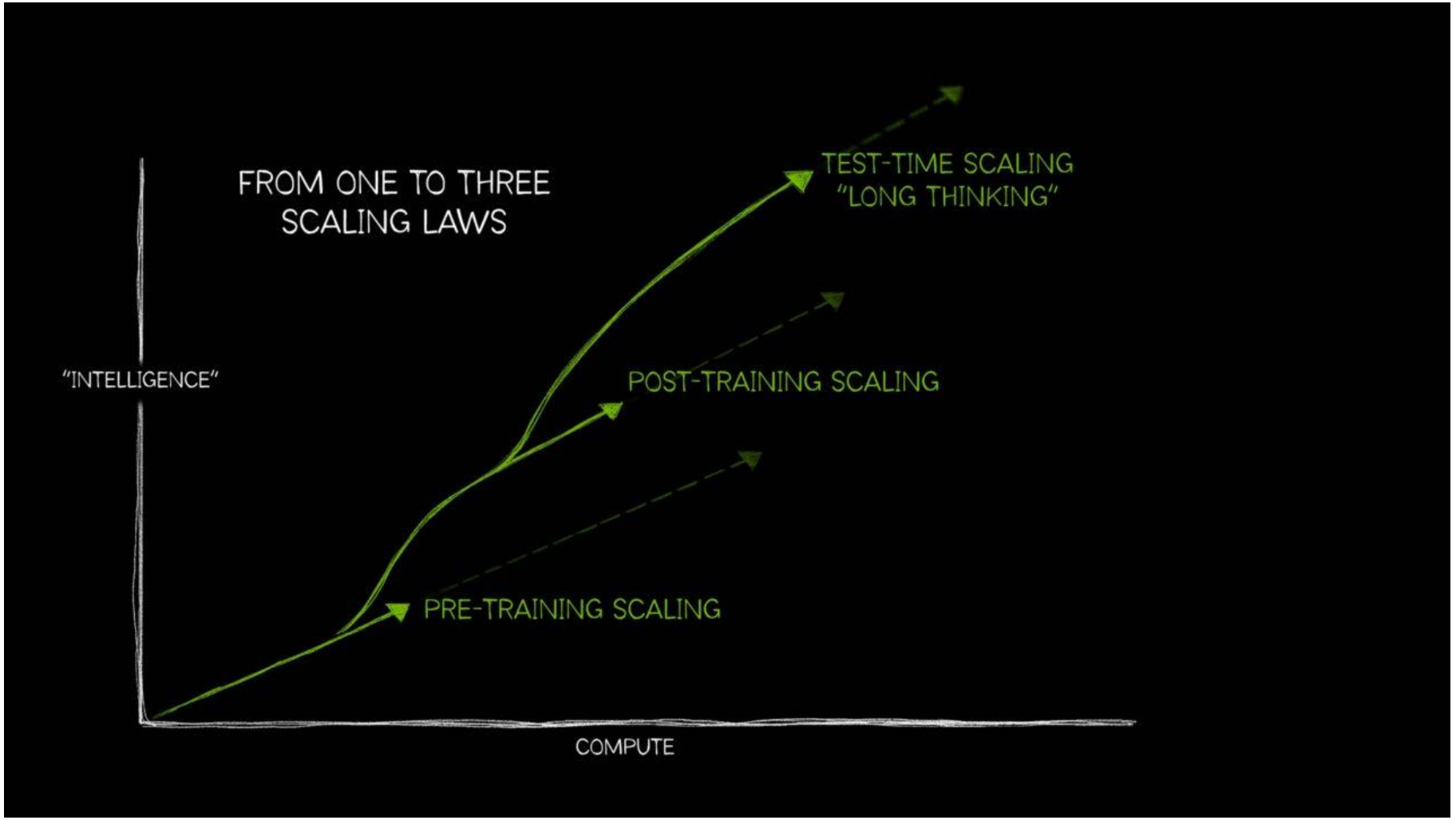


[Unleashing the Power of AI on Mobile: LLM Inference for Llama 3.2 Quantized Models with ExecuTorch and KleidiAI | PyTorch]

Inference Scenario 5: Augmented and Virtual Reality

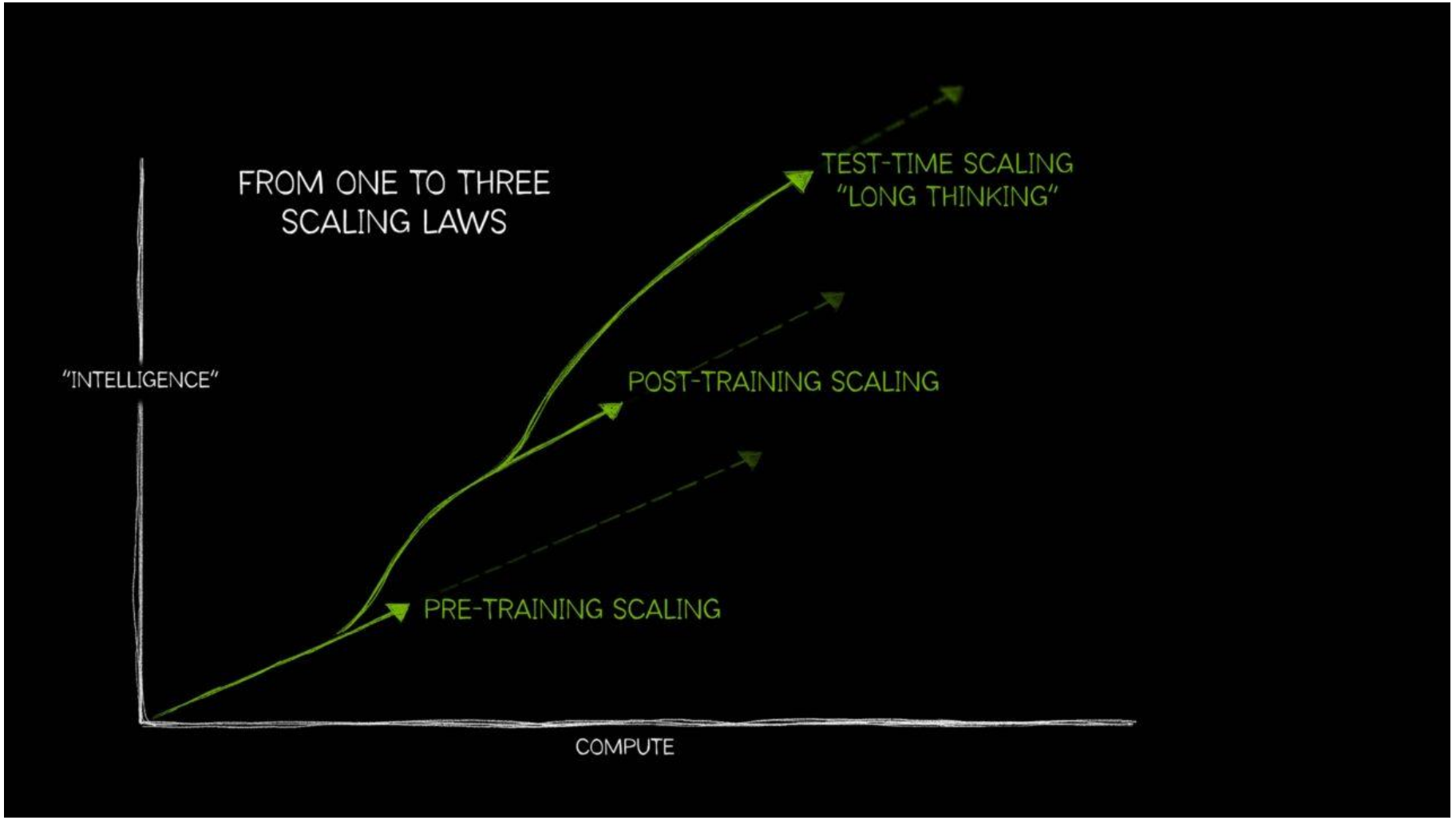


Inference Time Scaling



NVIDIA Jensen Huang at GTC 2025

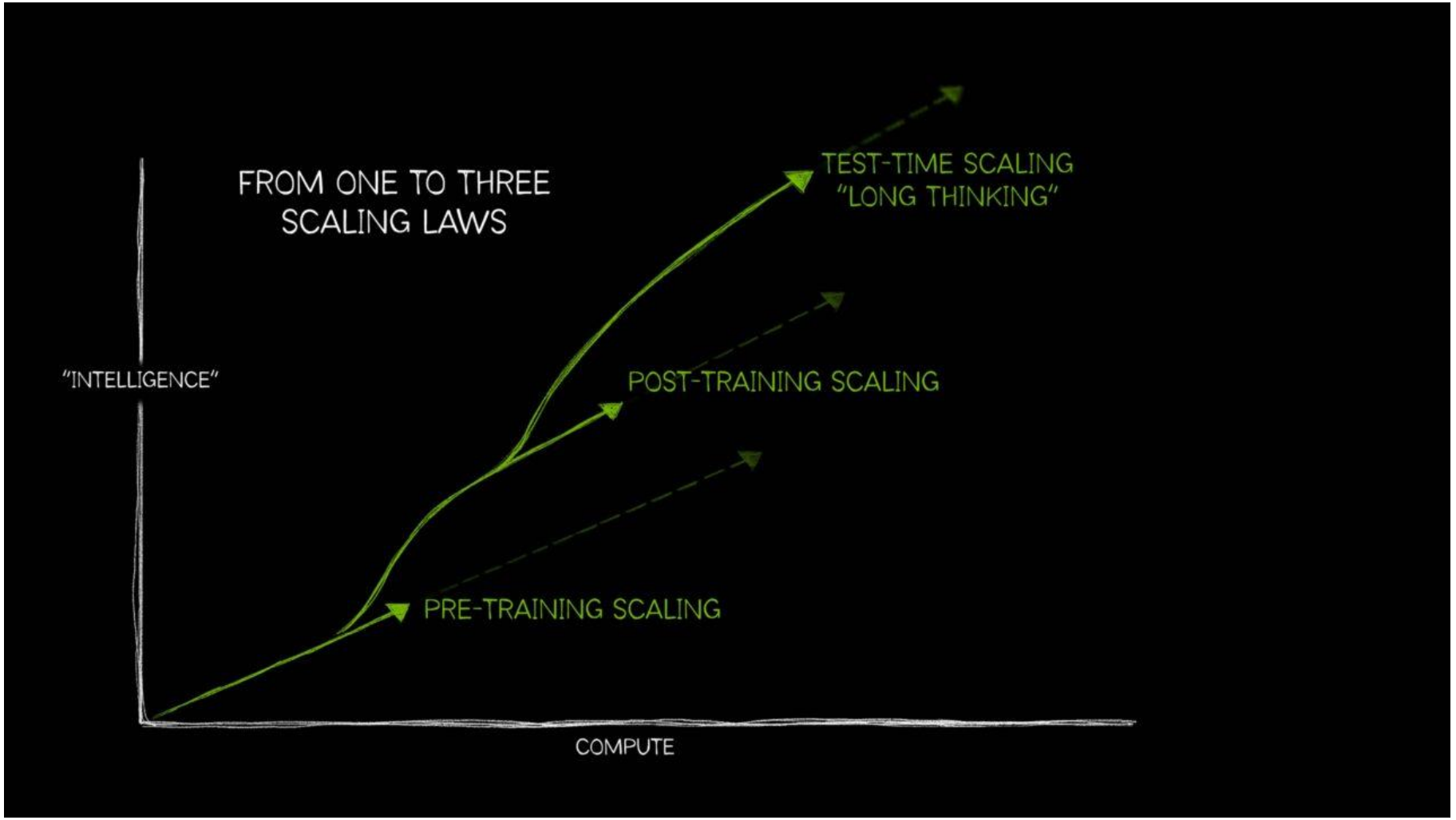
Inference Time Scaling



1 Pretraining scaling = bigger model and data, better performance

NVIDIA Jensen Huang at CES 2025

Inference Time Scaling

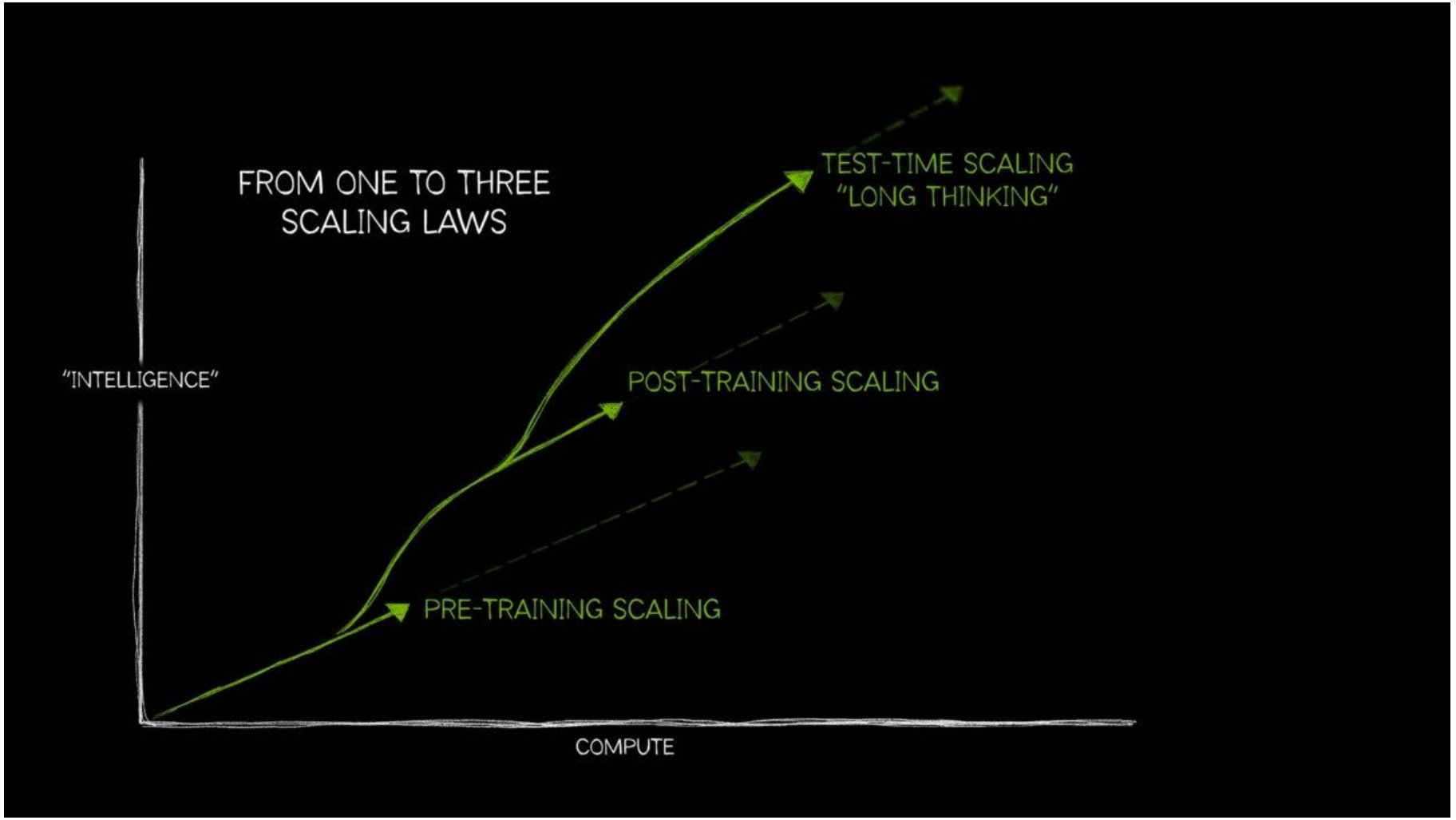


1 Pretraining scaling = bigger model and data, better performance

2 Post-training scaling = fine-tuning for precision

NVIDIA Jensen Huang at CES 2025

Inference Time Scaling Requires More Compute



1 Pretraining scaling = bigger model and data, better performance

2 Post-training scaling = fine-tuning for precision

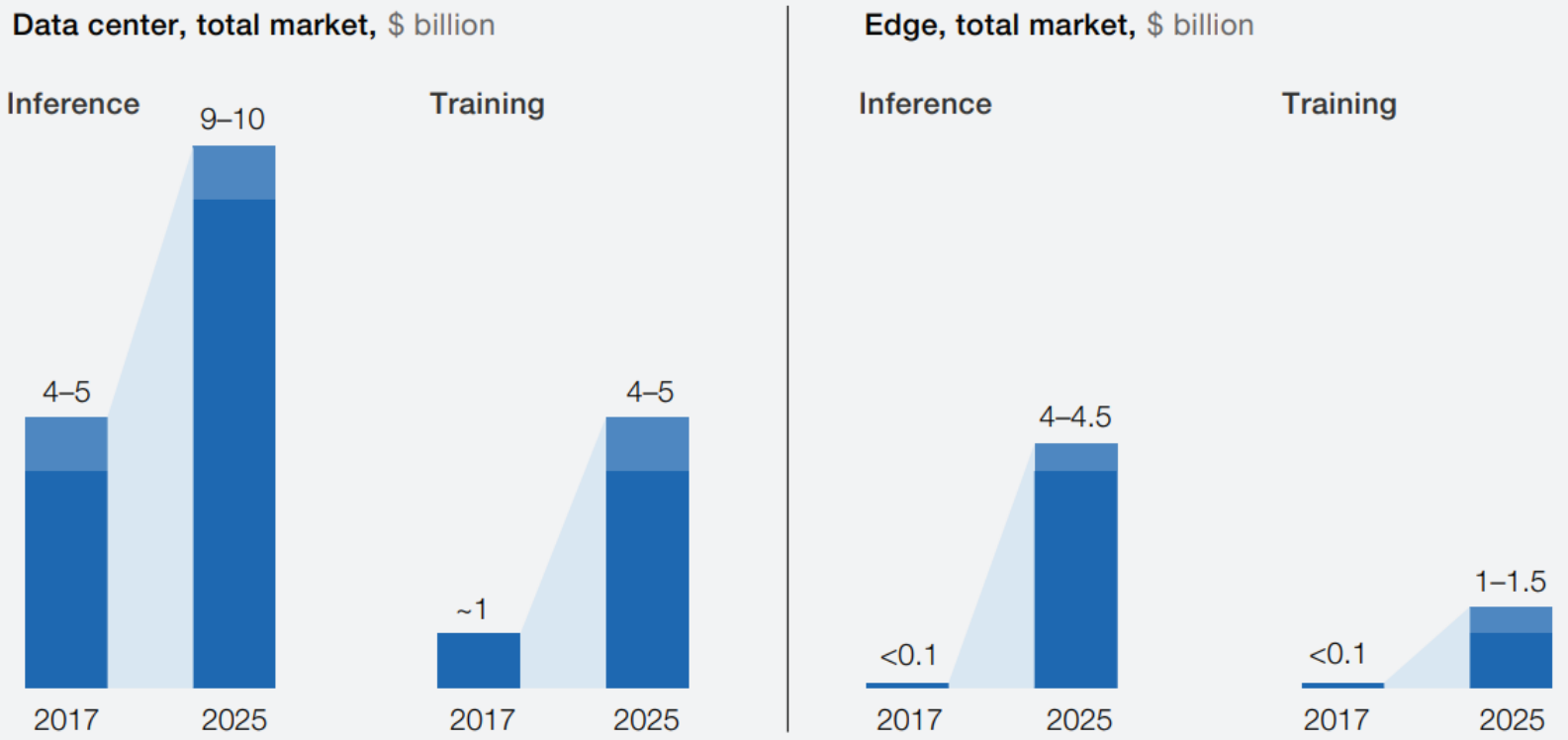
3 Test-time scaling (long thinking) = multi-pass reasoning for complex problems/try it many times

NVIDIA Jensen Huang at CES 2025

Training -> Inference

Exhibit 5

At both data centers and the edge, demand for training and inference hardware is growing.



Source: Expert interviews; McKinsey analysis

Training vs. Inference

	Training	vs	**Inference**
Runtime	Weeks or months		Milliseconds or seconds

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Challenges	Speed, Scale, Data		SLA (Service-level agreement), TCO (total cost of ownership)

Training vs. Inference

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SLA (LLM: token rates)

TCO

- Parameter volume
- LLM: Context length

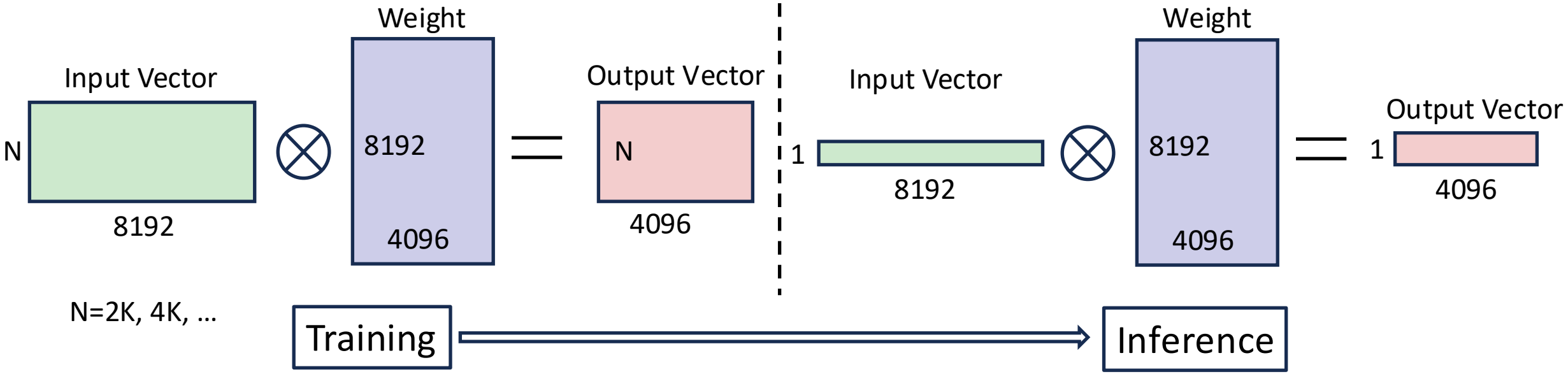
Inference Challenge 1: Long Latency Violates SLA

- Long serving latency blocks deployment
- Support advance models while meeting latency SLA and saving cost

DL Scenarios	Original Latency	Latency Target
Deep Query Document Similarity	10~12ms for [query, 1 doc] x 33 docs	< 6ms
Travel Planning	10ms for [query, 1 passage] x 150 passages	< 5ms
Ads seq2seq model for query rewriting	~51ms	< 5ms

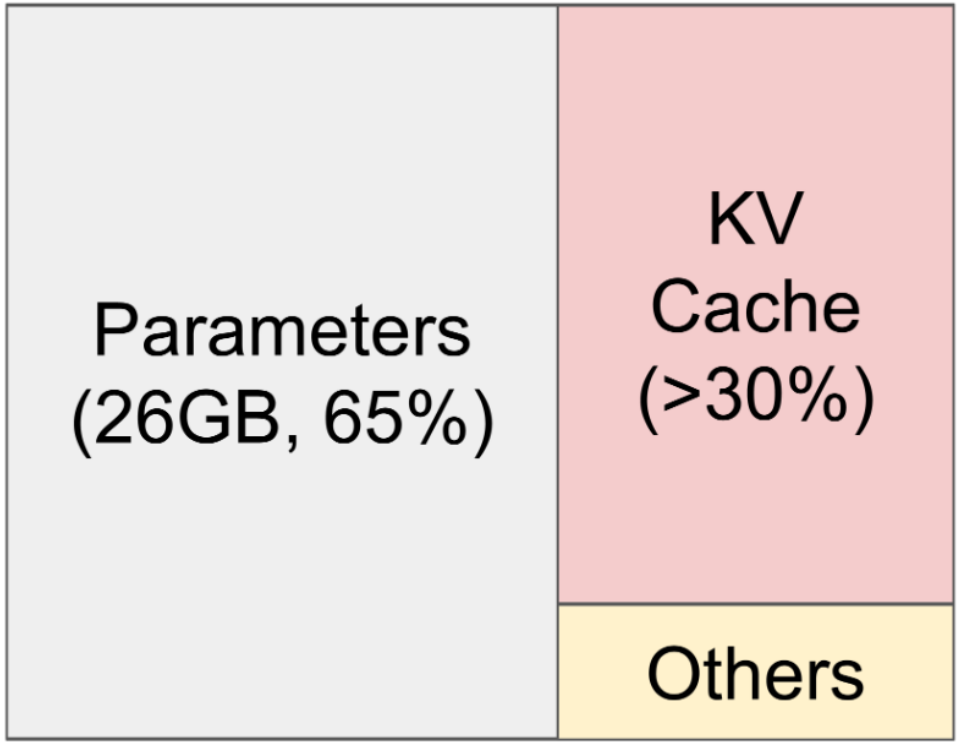
Inference Challenge 2: Small Batch Limits Parallelism

- Small batch size ⟹ Low latency yet low data reuse (arithmetic intensity)
- Autoregressive generation ⟹ Generate tokens one-by-one



Inference Challenge 3: Large Memory Increases Cost

- Model parameters
  - # Layers
  - # Hidden dim
- KV cache (in attention layer)
  - Batch size
  - Sequence length
  - # Layers
  - # Hidden
- Activation and others



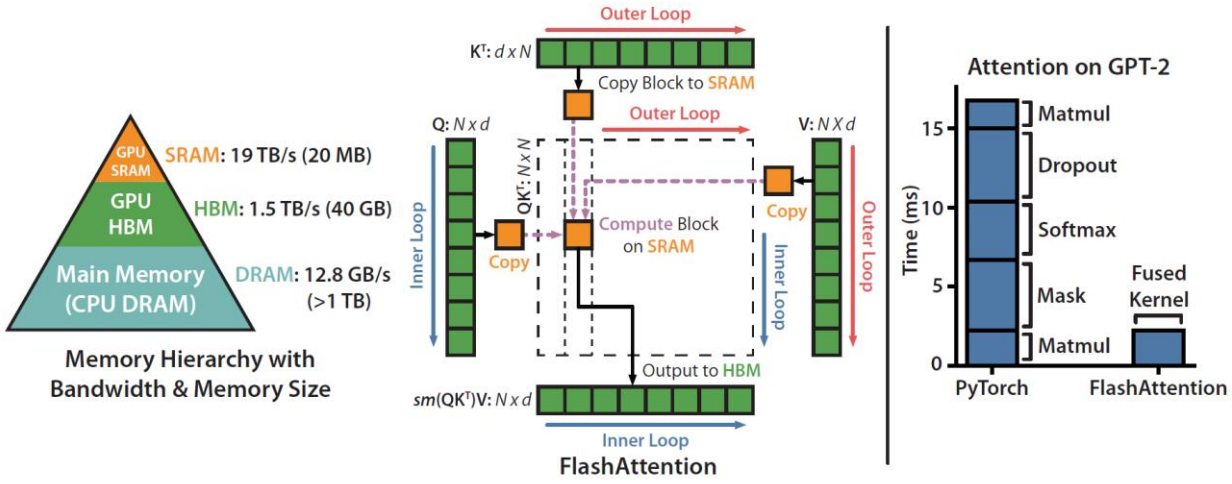
LLaMA-13B on A100 40 GB

Efficient Memory Management for Large Language Model Serving with PagedAttention, by Kwon et al., 2023

Recent Advances to Tackle These Inefficiencies

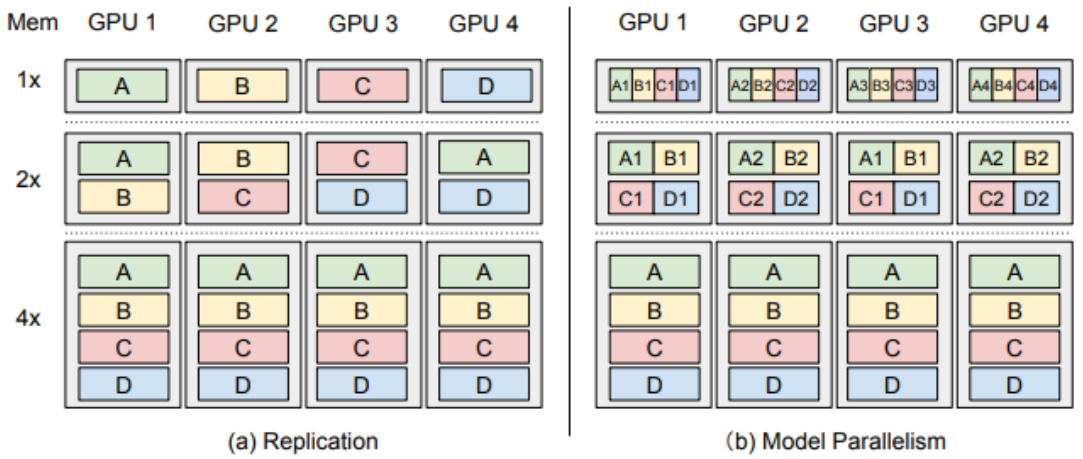
- From a down-to-top view
  - Execution layer (e.g., how to execute efficiently)
  - Planning layer (e.g., how to schedule resources)
  - Algorithm layer (e.g., how to change model)

Customized Kernels & Model Parallelism

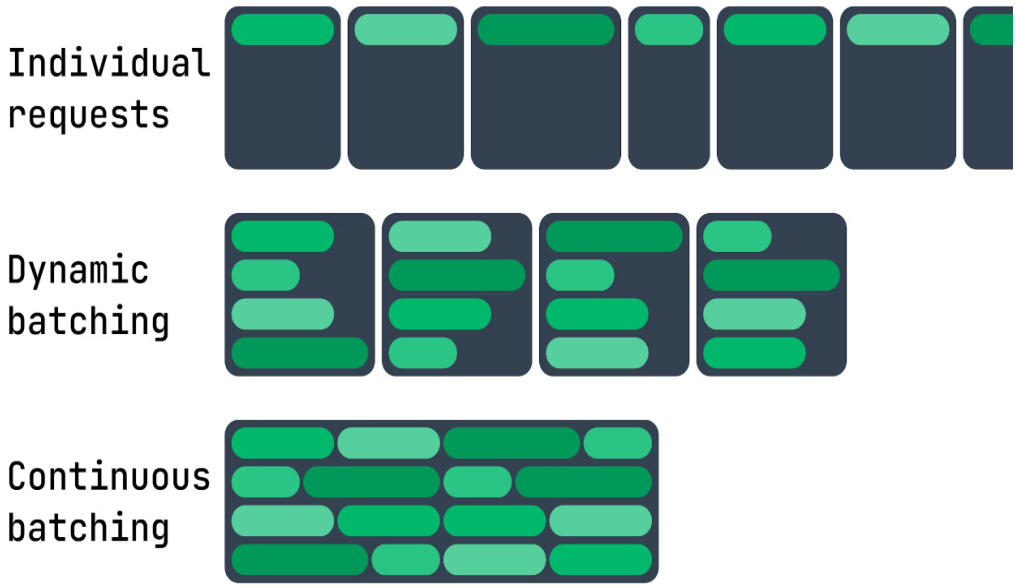


FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness, 2022

AlpaServe: Statistical Multiplexing with Model Parallelism for Deep Learning Serving, OSDI 2023

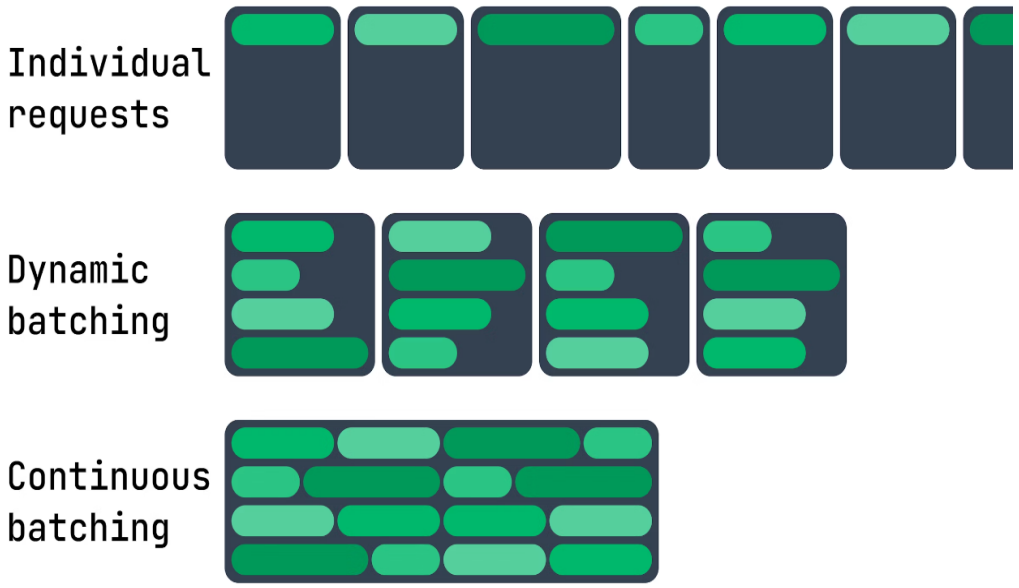


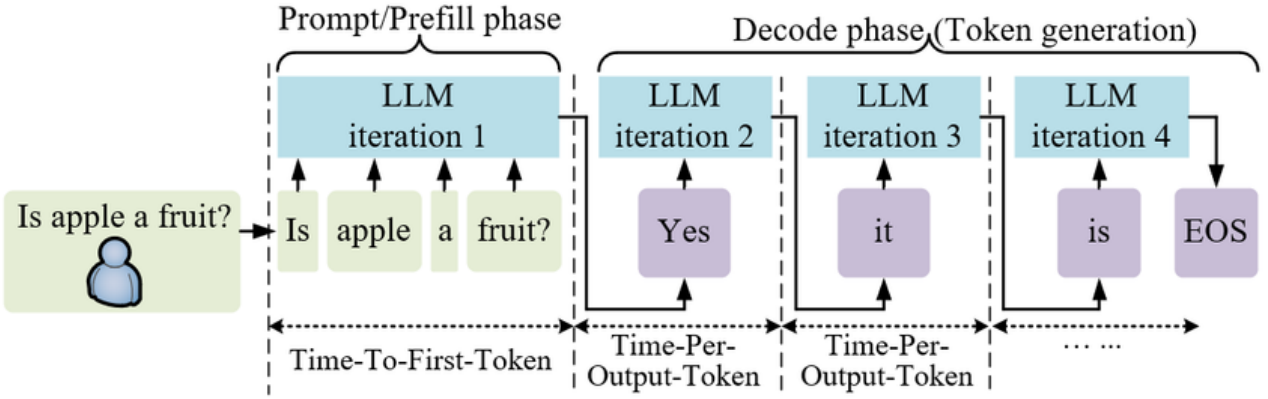
Scheduling Strategies for LLM Inference



Orca: A Distributed Serving System for Transformer-Based Generative Models, OSDI 2022

Scheduling Strategies for LLM Inference



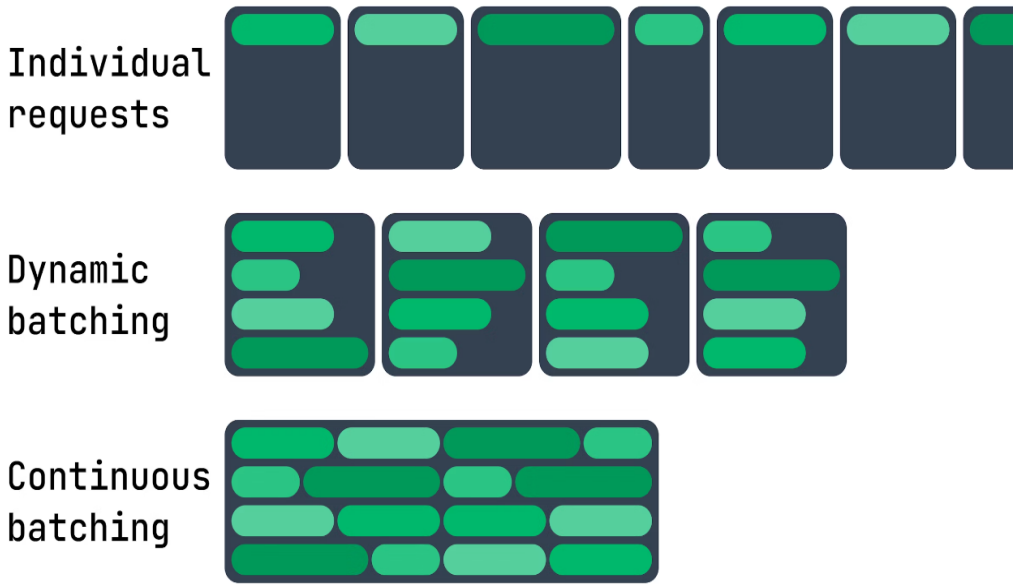


Orca: A Distributed Serving System for Transformer-Based Generative Models, OSDI 2022

LLM generation is autoregressive:

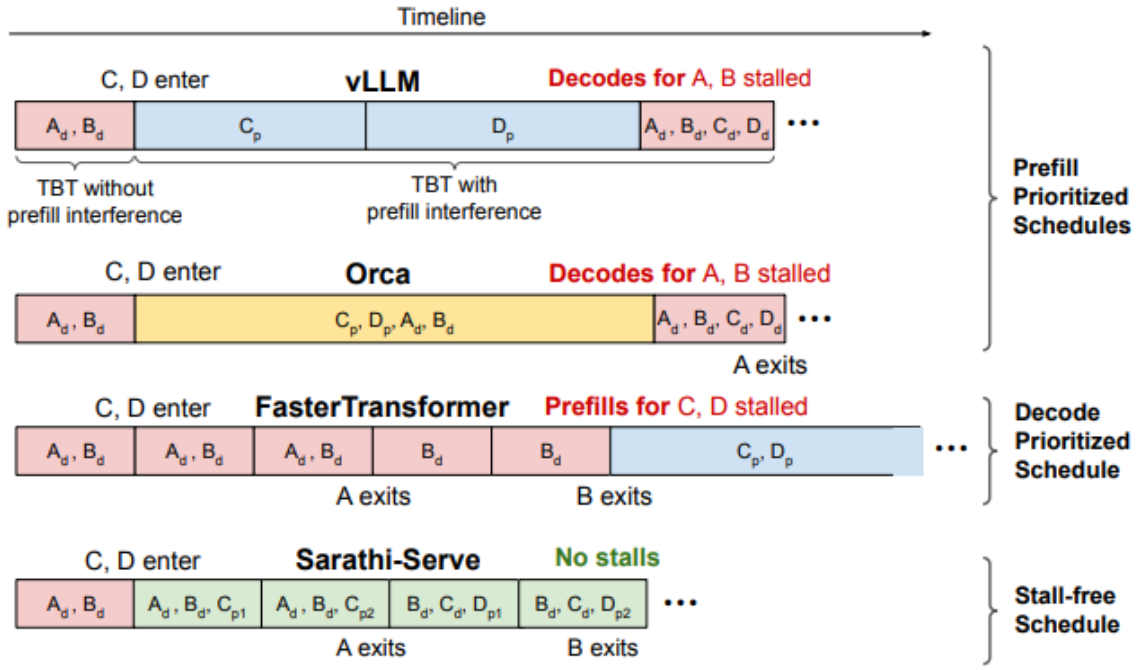
- Prefill stage: digest all input
- Decode stage: digest tokens by tokens

Scheduling Strategies for LLM Inference

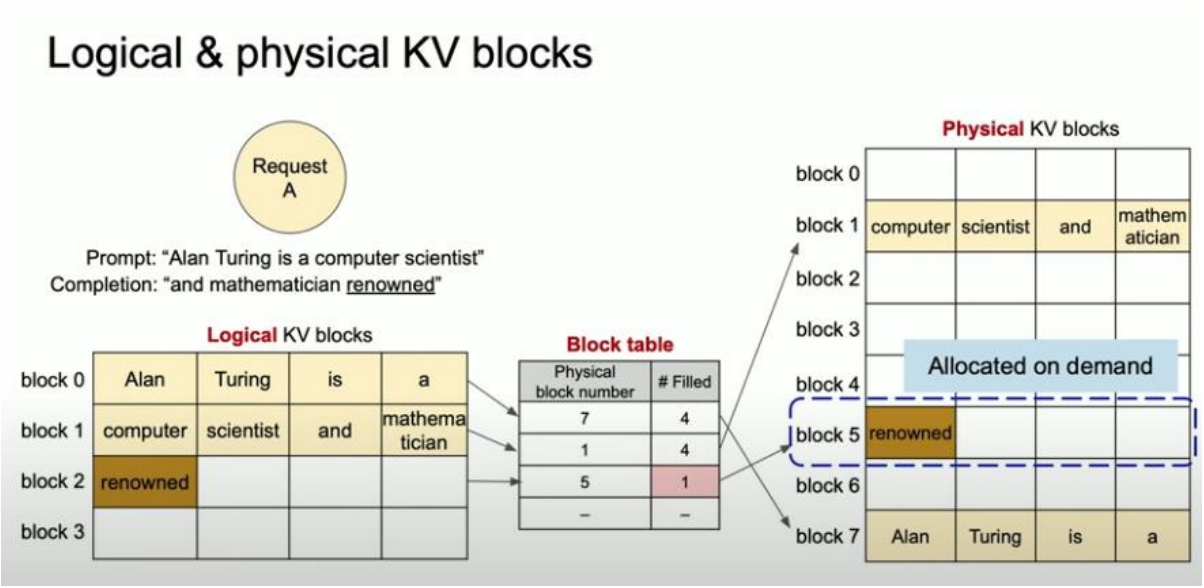


Orca: A Distributed Serving System for Transformer-Based Generative Models, OSDI 2022

Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve, OSDI 2024

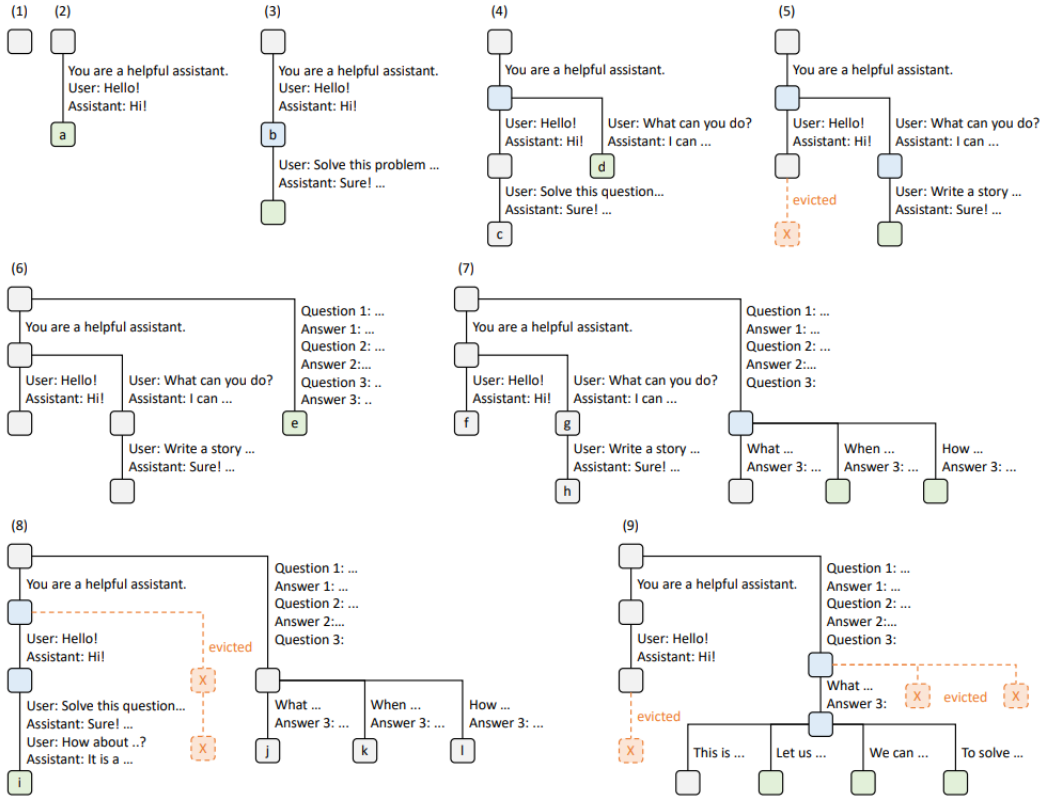


KV Cache Management



Efficient Memory Management for Large Language Model Serving with PagedAttention, 2023

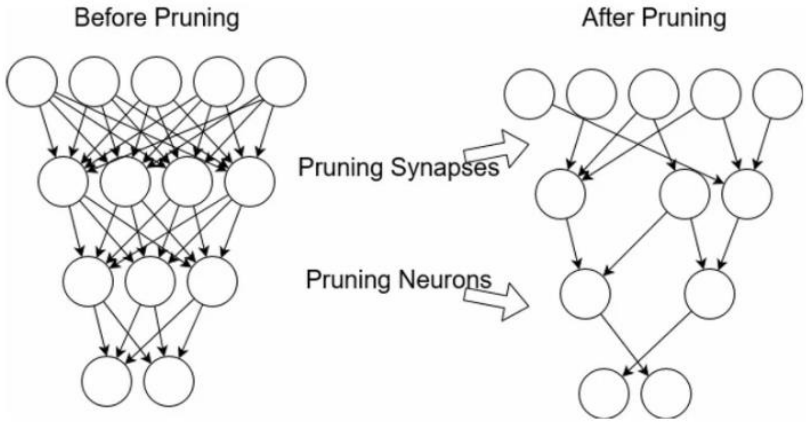
SGLang: Efficient Execution of Structured Language Model Programs, 2024



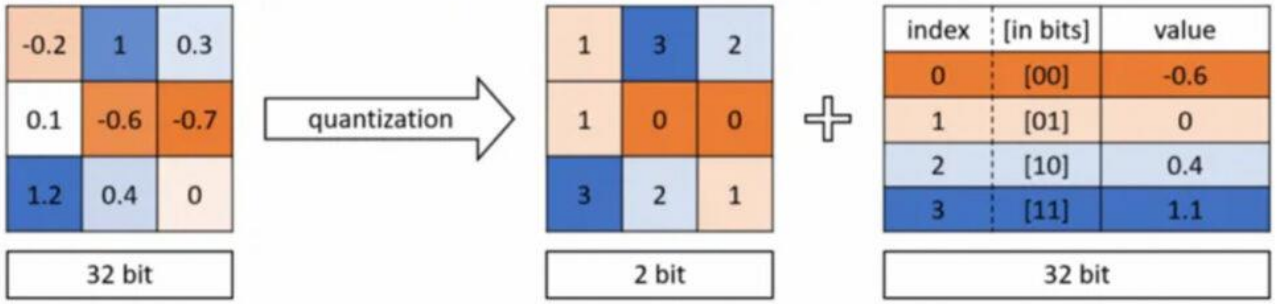
Recent Advances to Tackle These Inefficiencies

- From a down-to-top view
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Compression Strategies

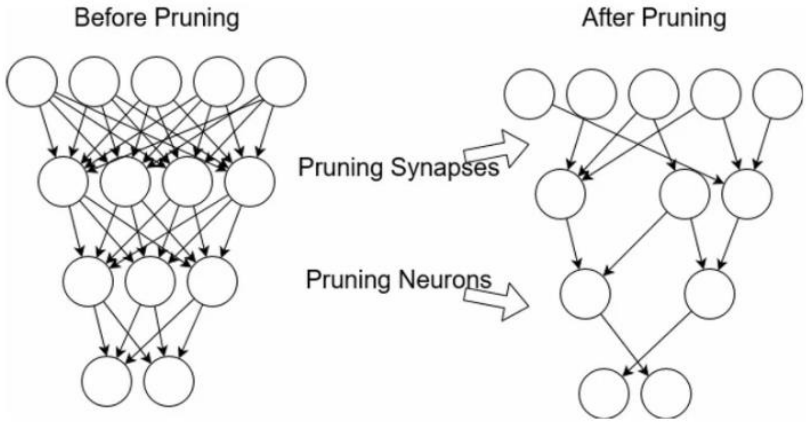


Pruning/Sparsification

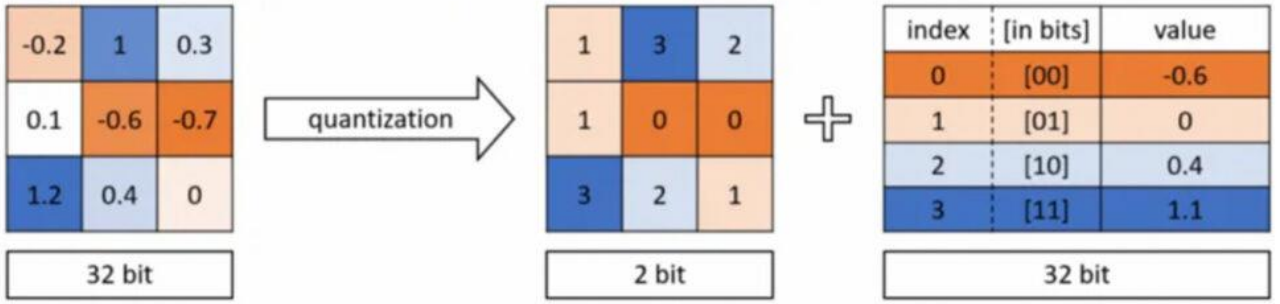


Quantization

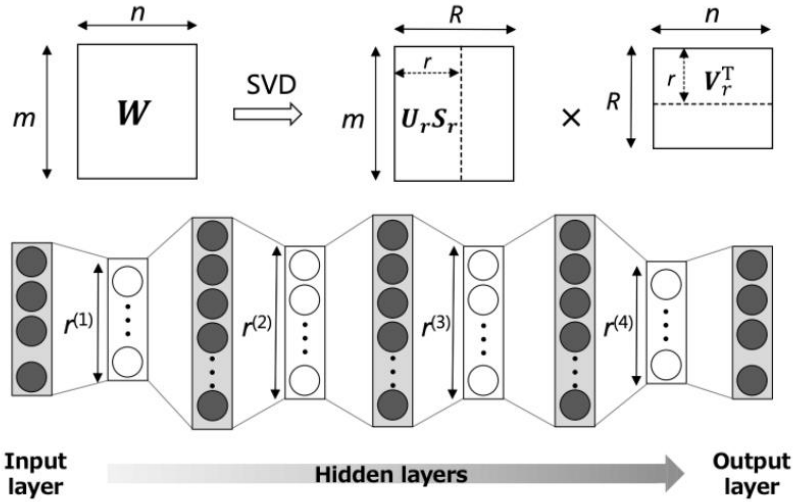
Compression Strategies



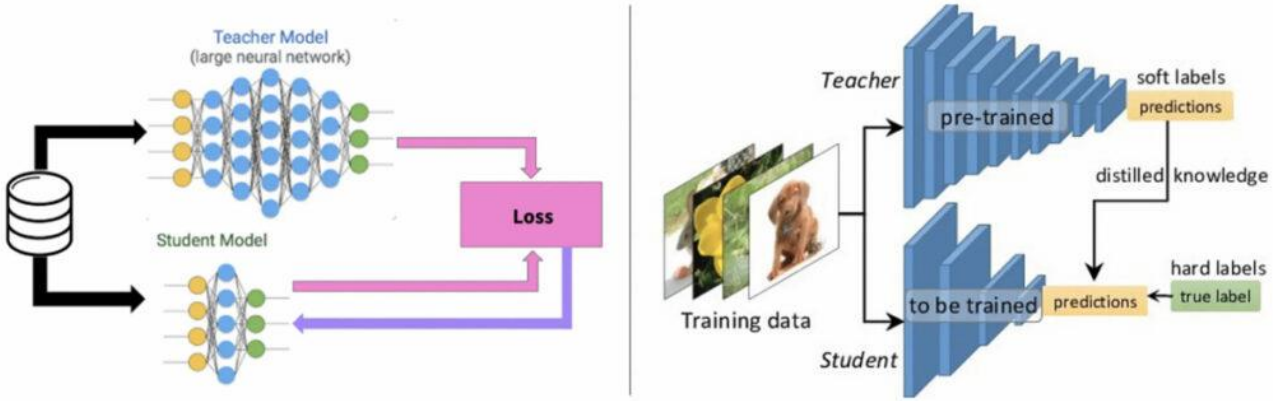
Pruning/Sparsification



Quantization



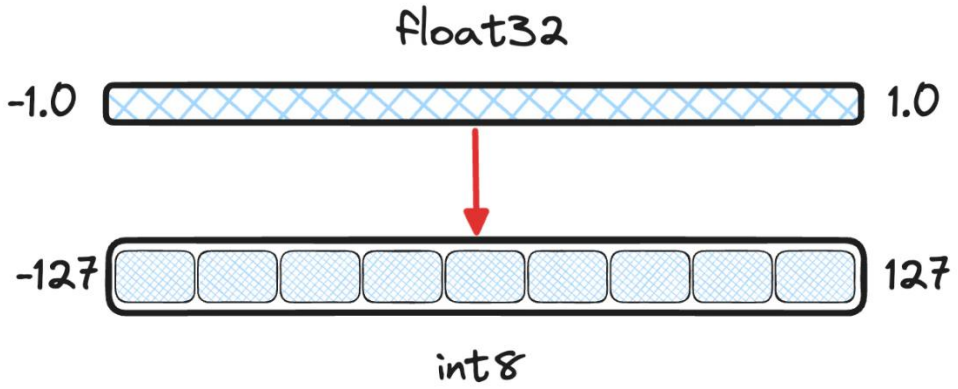
Low-rank decomposition



Distillation

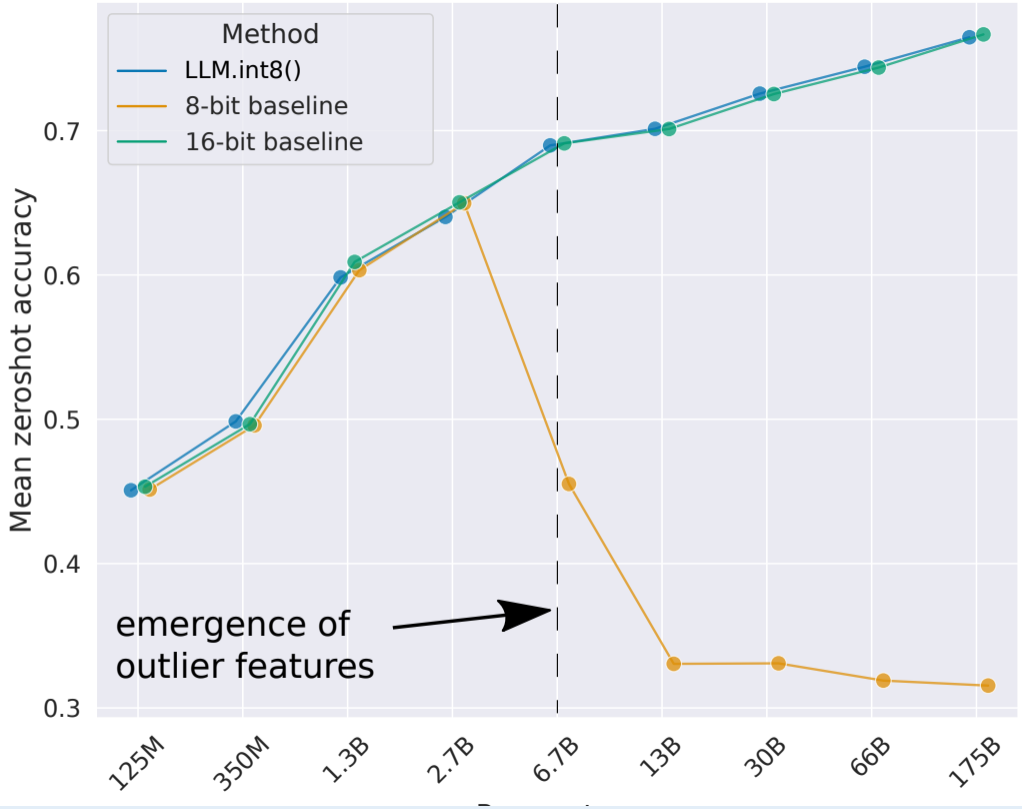
Quantization: Quick Recap

- Reduce the bits per weight, saving memory consumption
- Accelerate inference speed on supporting hardware



Challenges to Quantize LLMs

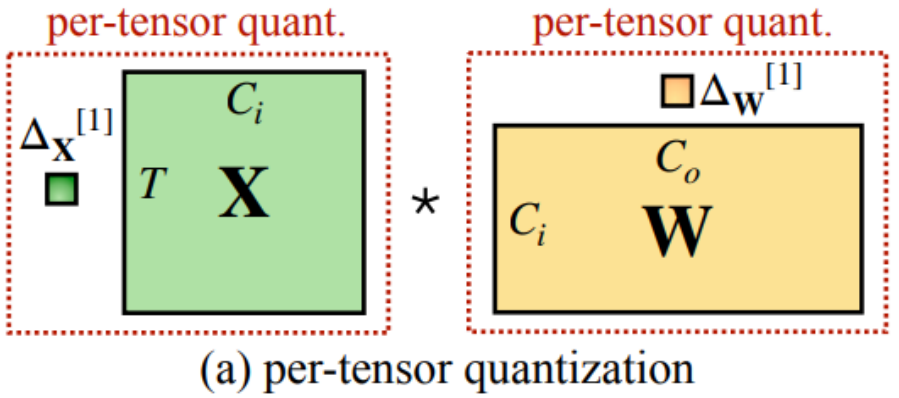
- Standard quantization strategy leads to catastrophic accuracy drop



LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale, 2023

Fine-grained Quantization

- Per-tensor quantization
  - Low accuracy
  - Fast to quantize/dequantize

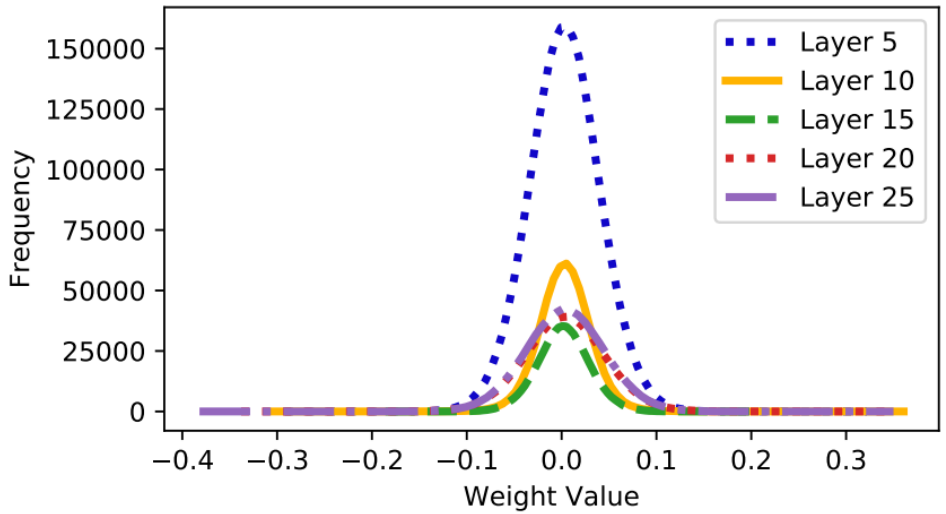


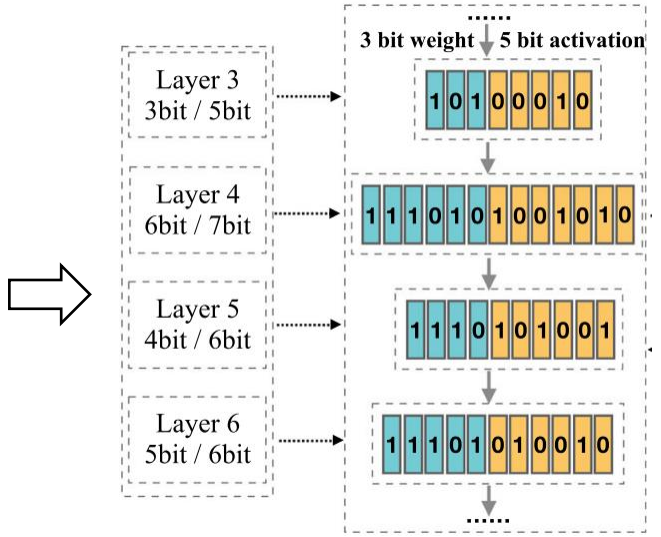
(a) per-tensor quantization

ZeroQuant: Efficient and Affordable Post-Training Quantization for Large-Scale Transformers, NeurIPS 2022

Mixed Precision Quantization

- Weights follow Gaussian distribution
- Outliers remain in original form, quantize the rest of the values
- Different bits for different layers



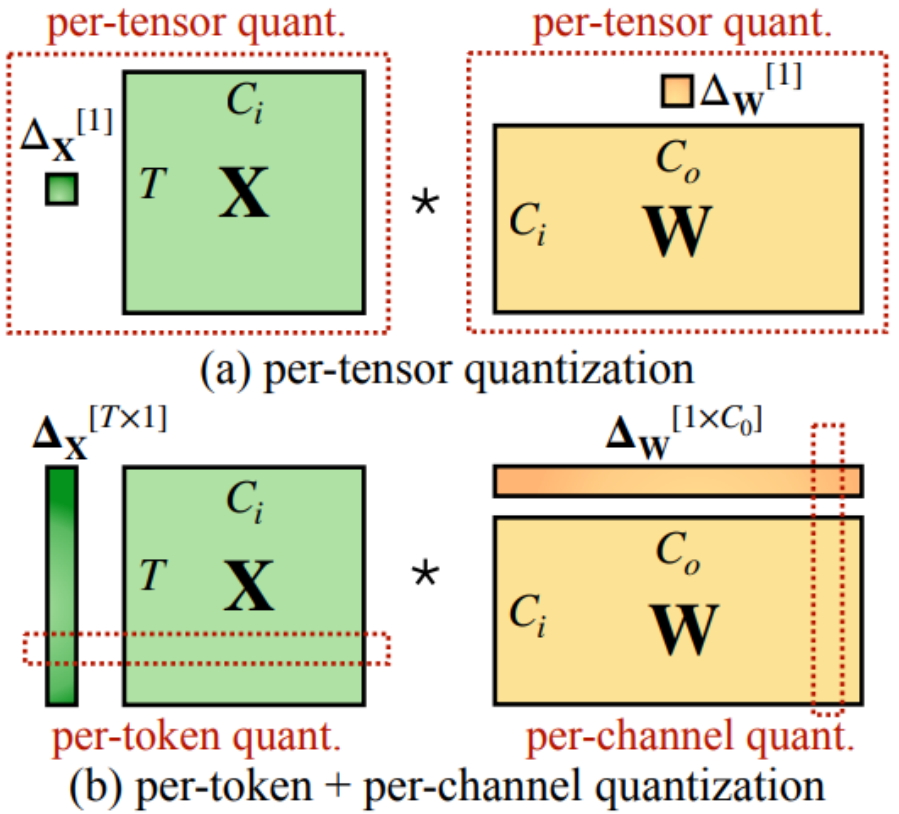


Per-layer weight distribution of BERT model

GOBO: Quantizing Attention-Based NLP Models for Low Latency and Energy Efficient Inference, MICRO 2020

Fine-grained Quantization

- Per-tensor quantization
  - Low accuracy
  - Fast to quantize/dequantize
- Per-token/channel quantization
  - High accuracy
  - Slower to quantize/dequantize
  - Custom kernels required

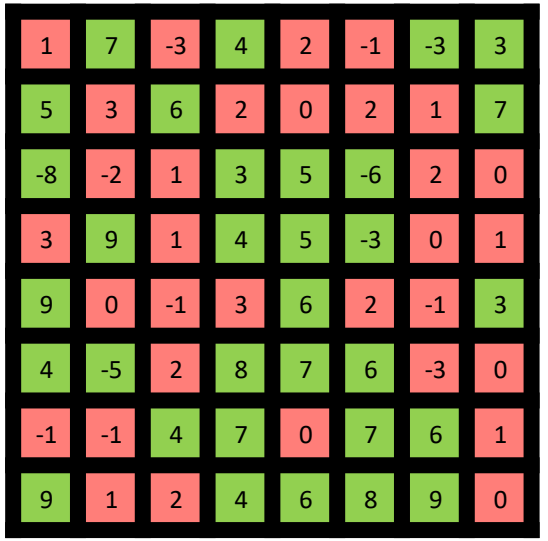


ZeroQuant: Efficient and Affordable Post-Training Quantization for Large-Scale Transformers, NeurIPS 2022

Sparsification

Unstructured (connection) Sparsity:

- High accuracy
- No performance improvement or performance regression



Unstructured Sparsity

SliceGPT: Compress Large Language Models by Deleting Rows and Columns, ICLR 2024

Sparsification

Unstructured (connection) Sparsity:

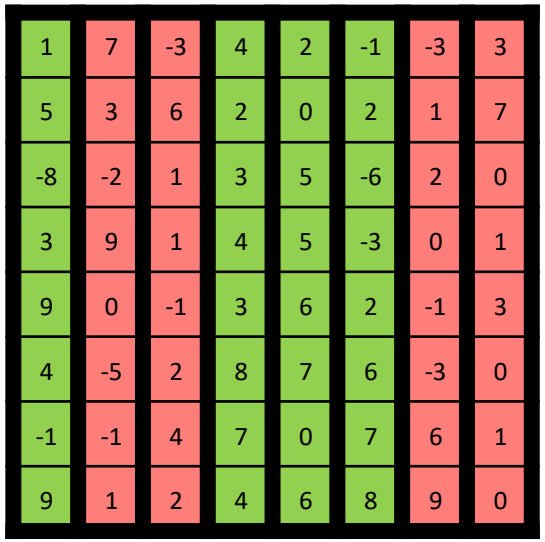
- High accuracy
- No performance improvement or performance regression

Structured Sparsity:

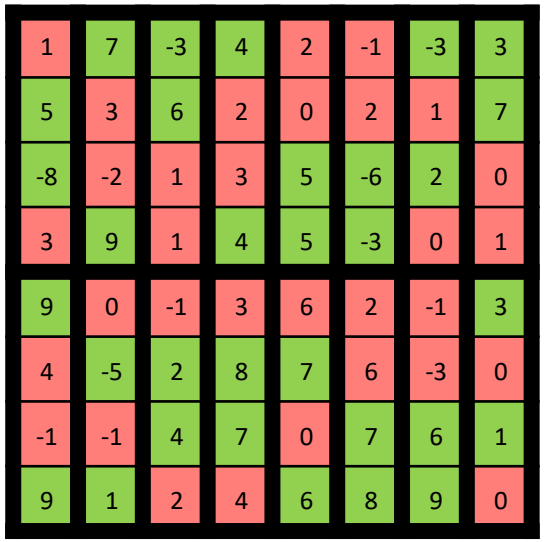
- Large accuracy degradation
- High performance scalability

N:M Semi-Structured Sparsity:

- High accuracy
- High performance improvement



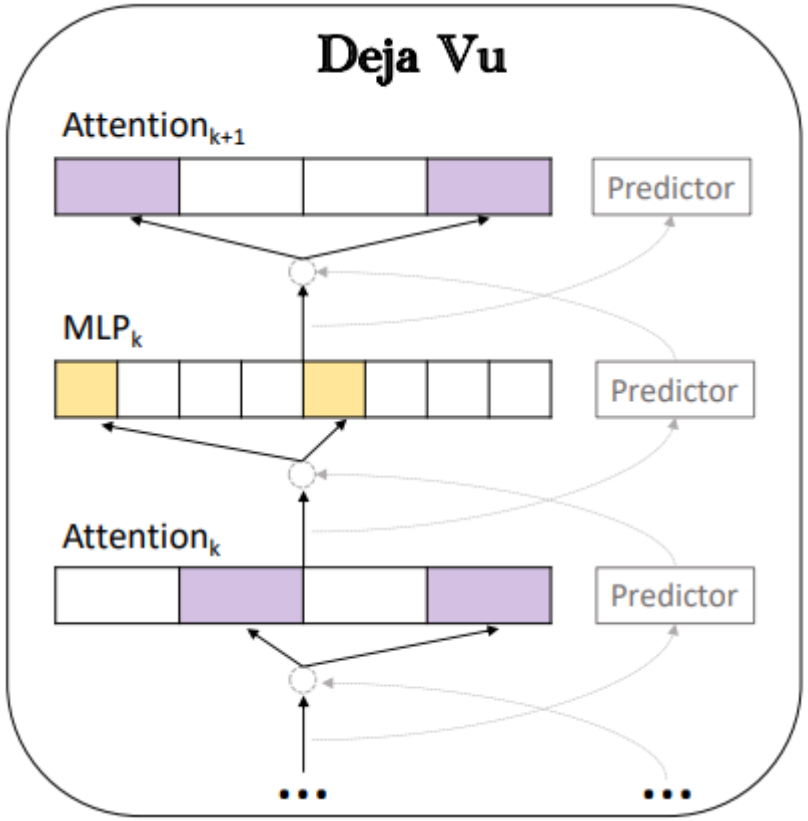
Structured Sparsity (Column-wise Sparsity)



Semi-Structured Sparsity (4:2 N:M)

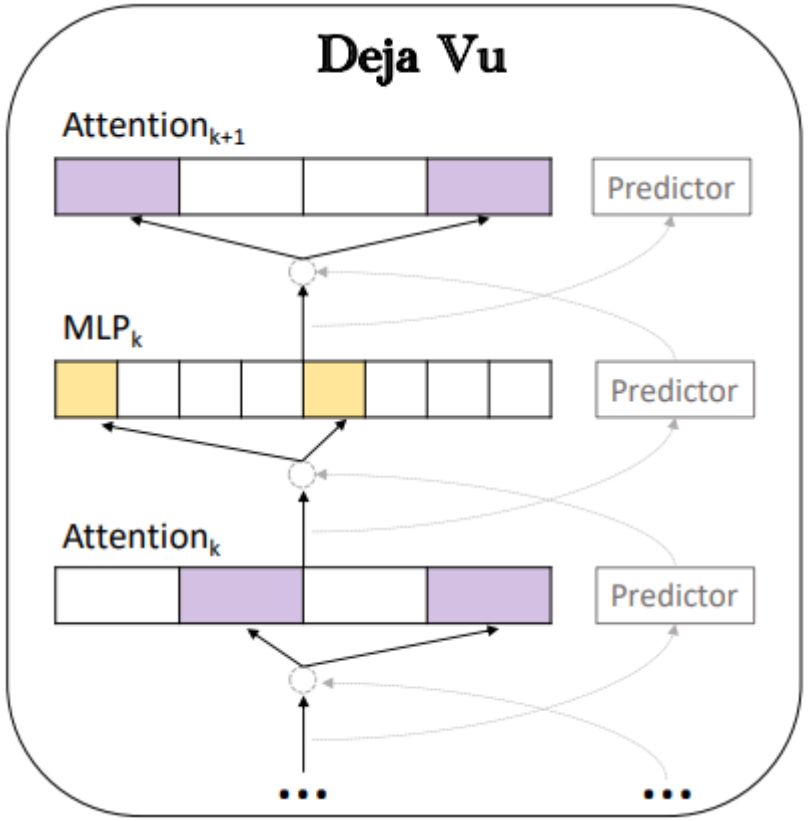
SliceGPT: Compress Large Language Models by Deleting Rows and Columns, ICLR 2024

Model Pruning



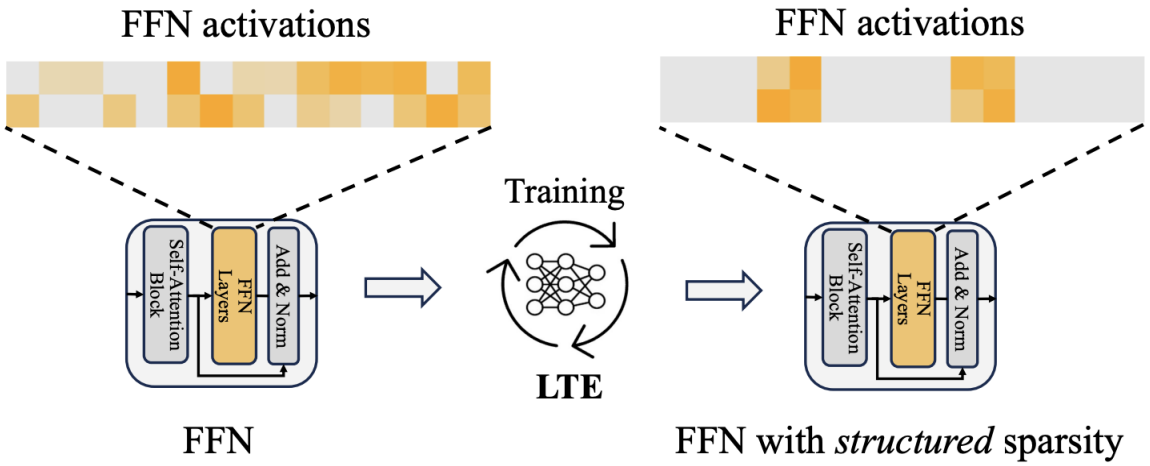
Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time, 2023

Model Pruning

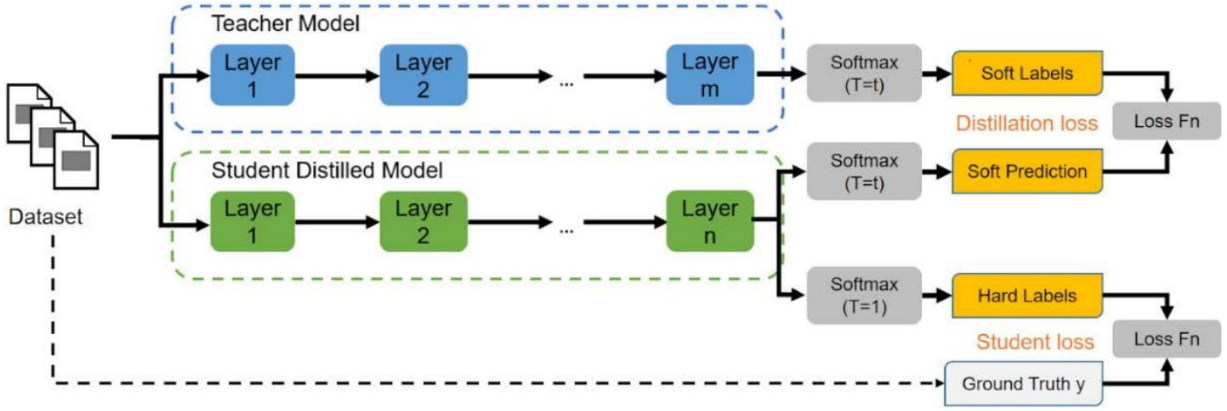


Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time, 2023

Learn To be Efficient: Build Structured Sparsity in Large Language Models, 2024

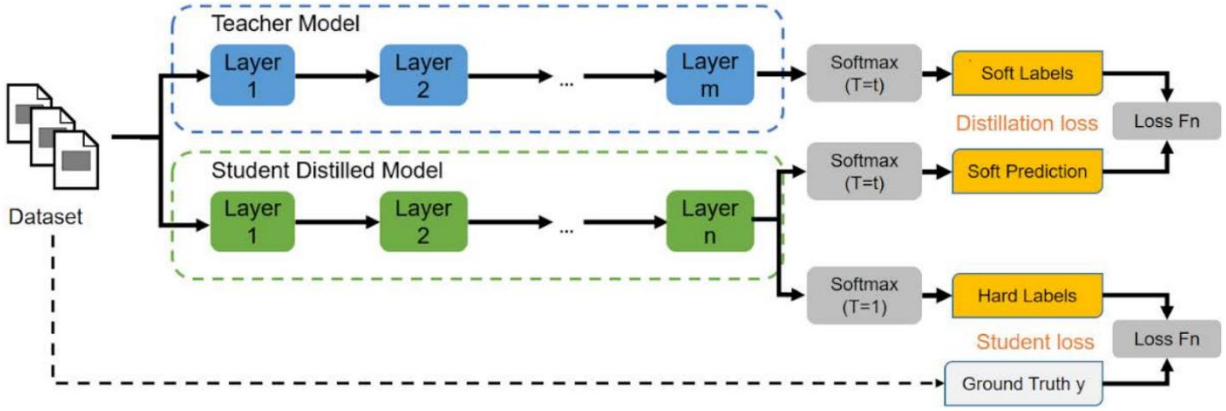


Knowledge Distillation



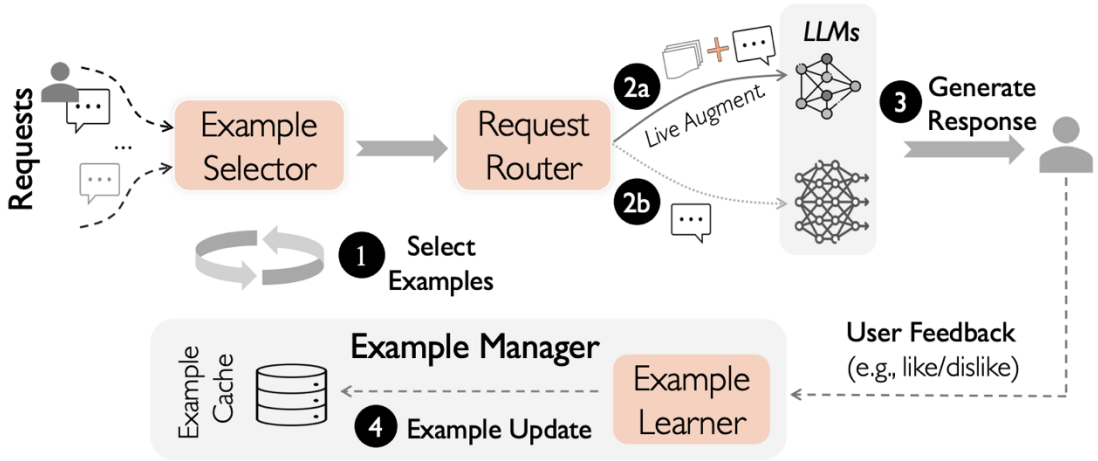
Distilling the Knowledge in a Neural Network , 2015

Knowledge Distillation



Distilling the Knowledge in a Neural Network , 2015

IC-Cache: Efficient Large Language Model Serving via In-context Caching, SOSP'25



Questions?